GERBE PROPERTY DESCENDS ALONG A 2-CARTESIAN SQUARE WITH LOCALLY ESSENTIALLY SURJECTIVE BASE LEG

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Gerbe criterion. lemma-when-gerbe (same chapter, named). Supplies properties (2)(a) and (2)(b) as the target conditions for X -> Y.
- Explicit 2-fibre product model. categories-lemma-2-product-categories-over-C (cross-chapter, named). Lets us assume X' = Y' x_Y X with explicit objects (U, y', x, alpha).

Source reference. Original reference 0040: focused-seeds/geometry/S9-stk-gerbe-descent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 06P1: sources/stacks-stacks/pinned-original-source.tex, source lines 2522–2522. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let*

$$\begin{array}{ccc} \mathcal{X}' & \xrightarrow{G'} & \mathcal{X} \\ {\scriptstyle F'}\downarrow & & \downarrow{\scriptstyle F} \\ \mathcal{Y}' & \xrightarrow{G} & \mathcal{Y} \end{array}$$

be a 2-cartesian diagram of stacks in groupoids over $\mathcal{C}$. If for every $U \in \mathrm{Ob}(\mathcal{C})$ and $x \in \mathrm{Ob}(\mathcal{Y}_U)$ there exists a covering $\{U_i \to U\}$ such that $x|_{U_i}$ is in the essential image of $G : \mathcal{Y}'_{U_i} \to \mathcal{Y}_{U_i}$ and $\mathcal{X}'$ is a gerbe over $\mathcal{Y}'$, then $\mathcal{X}$ is a gerbe over $\mathcal{Y}$.

Proof. By the uniqueness property of a 2-fibre product may assume that $\mathcal{X}' = \mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X}$ as in Categories, Lemma 0040. Let us prove properties (2)(a) and (2)(b) of Lemma 06P1 for $\mathcal{X} \to \mathcal{Y}$.

Let y be an object of $\mathcal{Y}$ lying over the object U of $\mathcal{C}$. By assumption there exists a covering $\{U_i \to U\}$ of U and objects $y'_i \in \mathcal{Y}'_{U_i}$ with $G(y'_i) \cong y|_{U_i}$. By (2)(a) for $\mathcal{X}' \to \mathcal{Y}'$ there exist coverings $\{U_{ij} \to U_i\}$ and objects x'_{ij} of $\mathcal{X}'$ over U_{ij} with $F'(x'_{ij})$ isomorphic to the restriction of y'_i to U_{ij}. Then $\{U_{ij} \to U\}$ is a covering of

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$\mathcal{C}$ and $G'(x'_{ij})$ are objects of $\mathcal{X}$ over U_{ij} whose images in $\mathcal{Y}$ are isomorphic to the restrictions $y|_{U_{ij}}$. This proves (2)(a) for $\mathcal{X} \to \mathcal{Y}$.

Let $U \in \mathrm{Ob}(\mathcal{C})$, let x_1, x_2 be objects of $\mathcal{X}$ over U, and let $b : F(x_1) \to F(x_2)$ be a morphism in $\mathcal{Y}_U$. By assumption we may choose a covering $\{U_i \to U\}$ and objects y'_i of $\mathcal{Y}'$ over U_i such that there exist isomorphisms $\alpha_i : G(y'_i) \to F(x_1)|_{U_i}$. Then we get objects

$$x'_{1i} = (U_i, y'_i, x_1|_{U_i}, \alpha_i) \quad \text{and} \quad x'_{2i} = (U_i, y'_i, x_2|_{U_i}, b|_{U_i} \circ \alpha_i)$$

of $\mathcal{X}'$ over U_i. The identity morphism on y'_i is a morphism $F'(x'_{1i}) \to F'(x'_{2i})$. By (2)(b) for $\mathcal{X}' \to \mathcal{Y}'$ there exist coverings $\{U_{ij} \to U_i\}$ and morphisms $a'_{ij} : x'_{1i}|_{U_{ij}} \to x'_{2i}|_{U_{ij}}$ such that $F'(a'_{ij}) = \mathrm{id}_{y'_i}|_{U_{ij}}$. Unwinding the definition of morphisms in $\mathcal{Y}' \times_{\mathcal{Y}} \mathcal{X}$ we see that $G'(a'_{ij}) : x_1|_{U_{ij}} \to x_2|_{U_{ij}}$ are the morphisms we're looking for, i.e., (2)(b) holds for $\mathcal{X} \to \mathcal{Y}$. $\square$